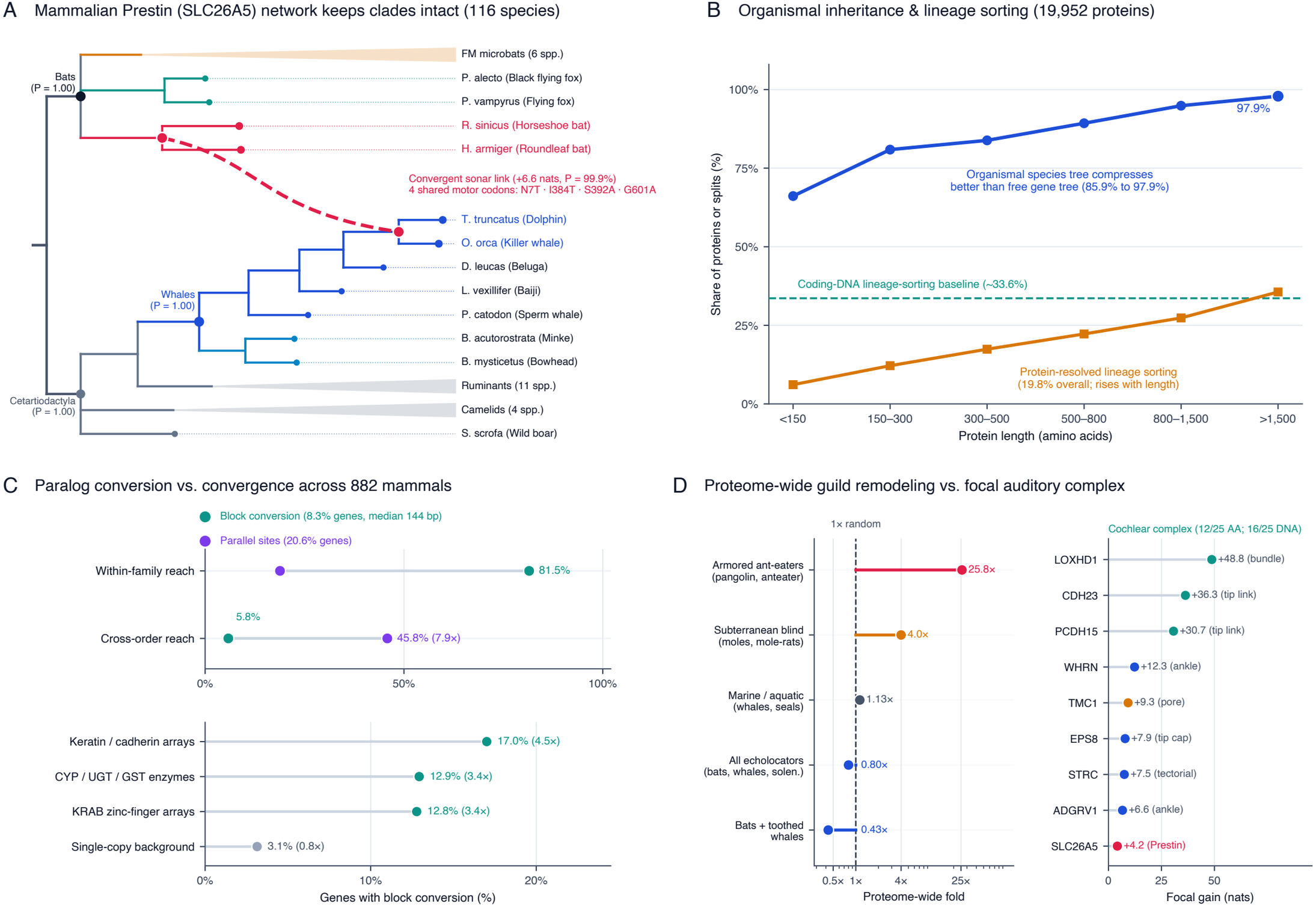
A
Mammalian Prestin (SLC26A5) network keeps clades intact (116 species)
Bats
(P = 1.00)
Whales
(P = 1.00)
Cetartiodactyla
(P = 1.00)
FM microbats (6 spp.)
P. alecto (Black flying fox)
P. vampyrus (Flying fox)
R. sinicus (Horseshoe bat)
H. armiger (Roundleaf bat)
Convergent sonar link (+6.6 nats, P = 99.9%)
4 shared motor codons: N7T · I384T · S392A · G601A
T. truncatus (Dolphin)
O. orca (Killer whale)
D. leucas (Beluga)
L. vexillifer (Baiji)
P. catodon (Sperm whale)
B. acutorostrata (Minke)
B. mysticetus (Bowhead)
Ruminants (11 spp.)
Camelids (4 spp.)
S. scrofa (Wild boar)
B
Organismal inheritance & lineage sorting (19,952 proteins)
Share of proteins or splits (%)
100%
75%
50%
25%
0%
97.9%
Organismal species tree compresses
better than free gene tree (85.9% to 97.9%)
Coding-DNA lineage-sorting baseline (~33.6%)
Protein-resolved lineage sorting
(19.8% overall; rises with length)
<150
150–300
300–500
500–800
800–1,500
>1,500
Protein length (amino acids)
C
Paralog conversion vs. convergence across 882 mammals
Block conversion (8.3% genes, median 144 bp)
Parallel sites (20.6% genes)
Within-family reach
81.5%
5.8%
Cross-order reach
45.8% (7.9×)
0%
50%
100%
Keratin / cadherin arrays
17.0% (4.5×)
CYP / UGT / GST enzymes
12.9% (3.4×)
KRAB zinc-finger arrays
12.8% (3.4×)
Single-copy background
3.1% (0.8×)
0%
10%
20%
Genes with block conversion (%)
D
Proteome-wide guild remodeling vs. focal auditory complex
1× random
Armored ant-eaters
(pangolin, anteater)
25.8×
Subterranean blind
(moles, mole-rats)
4.0×
Marine / aquatic
(whales, seals)
1.13×
All echolocators
(bats, whales, solen.)
0.80×
Bats + toothed
whales
0.43×
0.5×
1×
4×
25×
Proteome-wide fold
Cochlear complex (12/25 AA; 16/25 DNA)
LOXHD1
+48.8 (bundle)
CDH23
+36.3 (tip link)
PCDH15
+30.7 (tip link)
WHRN
+12.3 (ankle)
TMC1
+9.3 (pore)
EPS8
+7.9 (tip cap)
STRC
+7.5 (tectorial)
ADGRV1
+6.6 (ankle)
SLC26A5
+4.2 (Prestin)
0
25
50
Focal gain (nats)